

Same Referee, Different Games: Cross-Cube Evidence of Systematic Pipeline Bias in Simulated *Magic: The Gathering* Drafts

GIRY Aurélien

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Abstract

Bot-draft-and-simulate pipelines – pairing an automated drafting bot with an automated match engine – are an increasingly practical way to generate empirical, data-driven signal about a cube’s balance at a scale no human playtesting group can match. But such a pipeline is itself an instrument, not a neutral referee: if any of its stages carries a systematic bias, a single-cube study has no way to distinguish “this cube is unbalanced” from “this instrument has its own opinions.”

We test whether that concern is real by running one such pipeline (CubeCobra bot drafting and deck construction, Forge headless match simulation) against five independently designed, publicly available CubeCobra cubes spanning nearly the full power-level spectrum used in cube drafting – from an all-commons Pauper cube to the Power-9-inclusive MTGO Vintage Cube. A mono-color win-rate premium appears in all five, positive in direction throughout (+5.7 to +13.7 percentage points at the draft level) and reaching conventional statistical significance in three of the five when tested at the more conservative draft-paired level. A second, more specific pattern also recurs in all five: whatever mechanic in a given cube’s card pool requires the most sequential, precisely-timed planning to pay off – cantrip-based card selection in two cubes, a full storm-combo package in one, artifact-ramp-into-expensive-threats in two – has by far the lowest win rate in that cube, while efficient, low-commitment aggressive strategies dominate regardless of which specific cube is tested. As a further, more tentative observation, two of the five cubes happen to both include genuine cards with the always-on evasion keyword Shadow, despite no other overlap in card pool or design philosophy, and those cards outperform their own cube’s mono-color baseline in both. We conclude that a meaningful share of the card-level performance signal produced by this pipeline is systematic across cubes, rather than attributable to any individual cube’s design alone – while explicitly flagging that this study cannot yet localize that property to drafting, deck construction, or match play specifically.

1 Introduction

Cube designers increasingly have access to a genuinely useful empirical tool: pairing an automated drafting simulator with an automated match engine to generate large volumes of played-out games without needing a human playtest group. This makes data-driven cube design possible at a scale that was previously impractical – hundreds of drafts and thousands of matches can be generated and analyzed in a single session.

But this approach rests on an assumption worth checking explicitly: that the pipeline’s output reflects the cube’s actual balance and nothing else. Such a pipeline has several stages – an automated bot decides what to draft and how to build a deck from it, then an automated engine plays that deck out against others – and any one of those stages could carry its own systematic bias, independent

of what any particular card pool contains. A designer running this kind of pipeline on their own cube alone has no way to tell “my card pool is unbalanced” apart from “one stage of this pipeline has its own opinions.”

This paper tests that concern directly. We ran a bot-draft-and-simulate pipeline against five cubes, chosen deliberately for two properties: they are independently designed by different people with different goals and philosophies, and they are among the most-used cubes on CubeCobra, so any pattern found here is informative to a far larger audience than any single custom cube would be. If the same directional patterns turn up across five cubes that share essentially nothing – different card pools, different power levels, different design intentions – the most natural common factor to investigate is the one thing they were all run through: the pipeline itself. We deliberately do not claim to know *which* stage of the pipeline produces the effects reported below; §4 discusses why that question is left open, and what would be needed to answer it.

2 Methods

2.1 Cube selection

Table 1: The five cubes tested, spanning the power-level range used in cube drafting. All are public CubeCobra cubes; card pools and archetype labels are each cube’s own, unmodified.

Cube	Notes
The Pauper Cube	Commons only; the lowest power level tested.
World Championship Museum (96-04)	Built exclusively from cards that appeared in official World Championship decks, 1996–2004.
The 2017 Classic Modern Cube	Modern-legal power level.
The Peasant Cube 2026	Commons and uncommons.
MTGO Vintage Cube	Wizards’ official Vintage-format cube; includes the Power 9 and other Reserved List cards. The highest power level tested.

Cubes were selected for design independence (different curators, no shared lineage) and for being among CubeCobra’s highly drafted lists, so that this study concerns cube lists of practical relevance to existing CubeCobra users – we make no claim that five cubes constitute a representative sample of that user base, only that they are not an arbitrary or convenient one.

2.2 Pipeline stages

Each cube was drafted 50 times (400 decks) using CubeCobra’s bot-drafting simulator. Critically, *deck construction is also performed by this same bot* during the draft – CubeCobra’s drafting bot decides both what to pick and how to build its final maindeck/sideboard split, and this decision is exported directly as part of the draft record. Our pipeline converts that record verbatim into Forge’s .dck deck format: Forge is never asked to build a deck from a larger pool, and has no deck-construction role in this study. Forge’s only function here is playing pre-built decks against each other in a best-of-one, 8-seat round-robin (28 matches per draft, 1,400 matches per cube, 7,000 matches total across the five).

This matters for interpretation: the “pipeline” whose output we study bundles (a) CubeCobra’s bot-drafting policy, (b) CubeCobra’s bot deck-construction policy, and (c) Forge’s match-play policy

into a single measurement. All results below should be read as properties of this bundle, not as isolated to any one stage – see §4.1 for why we did not attempt to separate them in this study.

2.3 Operational definitions

A deck is classified as *mono-color* if its CubeCobra-assigned archetype label resolves to exactly one color letter (e.g. `W_Weenie`) and *multi-color* if it resolves to two or more (e.g. `RW_Aggro`); archetype labels that do not resolve to a recognizable color prefix (e.g. a generic `Combo` label) are excluded from this classification rather than forced into either bucket.

A card’s reported win rate is the pooled win rate of every deck that included it in its 40-card list for that match, *not* a rate conditioned on the card having actually been drawn, cast, or resolved in that game. This is a deliberate, necessary simplification – individual card-cast events are not tracked at that granularity by this pipeline – but it means a card’s reported win rate should be read as “how decks containing this card performed” rather than as an estimate of that card’s marginal, in-game impact. This distinction matters most for §3.2: a card’s win rate under this definition can be low either because it performs poorly when drawn, or because it tends to end up in decks that perform poorly for other reasons (an incomplete combo shell, for instance), and this paper’s data cannot separate the two.

A card is treated as *reliable* for win-rate reporting once it has appeared in at least two distinct drafts within a cube’s 50-draft corpus, to avoid the case where a card seen in only one deck mechanically inherits that deck’s overall win/loss record rather than providing an independent signal; Tables 4 and 5 report only reliable cards. We checked this threshold’s effect on §3.2’s results by recomputing Table 4 at ≥ 5 and ≥ 10 distinct drafts instead of ≥ 2 : the top-15/bottom-15 averages shift by at most 2 percentage points in any cube, and by nothing at all in three of the five, so the pattern reported there is not an artifact of the more permissive threshold used in the main tables. The “bottom-15 characterized by” column in Table 4 reflects the authors’ manual, qualitative reading of what those fifteen cards have in common, produced after observing the ranking – it is an exploratory characterization, not a pre-defined or quantified category (see §4.4).

2.4 Guardrails

Two measurement guardrails were established and validated before this study, against manual inspection of raw match logs. First, win detection is anchored to the single, unambiguous `Match Winner` – log line rather than pattern-matching narrative outcome text, which is vulnerable to AI turn-timeouts emitting contradictory outcome lines for the same game. Second, the legal blocking rate (used only in supporting analysis, not in the results reported here) is computed from a reconstructed board state rather than a naive count of “didn’t block” lines, which conflates an empty board, a legally-unblockable attacker, and a genuine missed block. No card list, keyword list, or analysis parameter was adjusted for any individual cube tested here.

2.5 Statistical tests

The primary test for §3.1 is a one-sample, two-tailed t -test and a Wilcoxon signed-rank test (both against $H_0 : \Delta = 0$) on the per-draft $\Delta = WR_{\text{mono}} - WR_{\text{multi}}$ values within each cube; 95% confidence intervals on the mean use the t -distribution. The pooled match-level comparison in Table 3 uses a 2-proportion z -test, reported for continuity with an earlier version of this analysis and discussed critically in §3.1 and §4.3 rather than treated as primary evidence.

2.6 Software versions

All matches were played using Forge 2.0.14 (`forge-installer-2.0.14`). All five cubes’ draft data was exported from CubeCobra’s bot-drafting simulator on 15 August 2026; any changes to CubeCobra’s bot-drafting policy after that date are not reflected in this study.

3 Results

3.1 A mono-color premium, confirmed at the draft level in three of five cubes

The primary test of the mono-color premium reported here is a draft-level paired comparison: for each draft, $\Delta = WR_{\text{mono}} - WR_{\text{multi}}$ (the pooled mono-color win rate minus the pooled multi-color win rate within that one draft), giving up to 50 approximately independent observations per cube rather than 1,400 non-independent match outcomes. Drafts with zero mono-color or zero multi-color decks (no basis for a within-draft comparison) are excluded; this was rare in four of the five cubes but notable in the MTGO Vintage Cube, where 16 of 50 drafts (32%) had no directly comparable mono-color deck, itself an observation about that cube’s drafting environment worth noting in passing.

Table 2: Draft-level paired comparison, $\Delta = WR_{\text{mono}} - WR_{\text{multi}}$ per draft. One-sample t -test and Wilcoxon signed-rank test against $H_0 : \Delta = 0$; 95% CI on the mean via the t -distribution.

Cube	n drafts	Mean Δ	95% CI	% drafts > 0	t -test p	Wilcoxon p
The Pauper Cube	49	+13.66	[8.31, 19.02]	75.5%	5.2×10^{-6}	2.5×10^{-5}
World Championship Museum	49	+7.97	[1.98, 13.97]	59.2%	1.0×10^{-2}	8.3×10^{-3}
The 2017 Classic Modern Cube	47	+8.56	[3.14, 13.98]	66.0%	2.6×10^{-3}	1.6×10^{-3}
The Peasant Cube 2026	48	+5.68	[-0.87, 12.24]	58.3%	8.8×10^{-2}	8.7×10^{-2}
MTGO Vintage Cube	34	+7.45	[-1.13, 16.03]	64.7%	8.7×10^{-2}	9.1×10^{-2}

At this more conservative unit of analysis, the point estimate is positive in all five cubes, and reaches conventional significance ($p < 0.05$, both tests agreeing closely) in three: the Pauper Cube, World Championship Museum, and the 2017 Classic Modern Cube. In the remaining two – the Peasant Cube 2026 and the MTGO Vintage Cube – the 95% confidence interval crosses zero, and a majority of individual drafts still favor mono-color (58% and 65% respectively), but the result does not reach significance at this sample size. We report this honestly as a weaker and more mixed result than the pooled match-level test below suggests, rather than treating the three significant cubes as representative of all five.

For comparison, and because it was our original (less rigorous) test, the pooled match-level result is reproduced in Table 3. It agrees in direction with Table 2 in every cube, but its 2-proportion z -test treats each match as an independent observation, which §4.3 discusses in more detail; we no longer treat its p -values as the primary evidence for this finding; and this discrepancy from the previous version of this analysis is itself the reason we no longer do so.

Three of five individually significant leaves open a natural question: is the effect itself, combined across all five independently designed cubes, distinguishable from zero? A fixed-effect, inverse-variance-weighted meta-analysis of the five per-cube estimates in Table 2 answers this directly: pooled $\Delta = +9.15$ points, $SE = 1.36$, 95% CI [6.49, 11.81], $z = 6.74$, $p = 1.6 \times 10^{-11}$. A random-effects (DerSimonian–Laird) version of the same meta-analysis, run as a sensitivity check on the fixed-effect assumption, gives an almost identical estimate ($\Delta = +9.11$, $SE = 1.43$, 95% CI [6.32, 11.90], $p = 1.7 \times 10^{-10}$), which is reassuring but not dispositive given only five studies to es-

Table 3: Pooled mono- vs. multi-color win rate, all five cubes (match-level, non-independent observations – see discussion above and §4.3).

Cube	Mono WR	Multi WR	Gap (pts)	z	p
The Pauper Cube	56.90%	44.69%	+12.21	6.40	1.5×10^{-10}
World Championship Museum	53.91%	46.32%	+7.59	3.98	6.9×10^{-5}
The 2017 Classic Modern Cube	55.48%	47.76%	+7.72	3.81	1.4×10^{-4}
The Peasant Cube 2026	54.98%	48.36%	+6.62	3.02	2.5×10^{-3}
MTGO Vintage Cube	55.43%	49.22%	+6.20	2.17	3.0×10^{-2}

timate between-cube variance from. A Cochran’s Q test for heterogeneity across the five cube-level estimates does not reject homogeneity ($Q = 4.37$, $df = 4$, $p = 0.36$); we read this as finding no evidence of substantial heterogeneity among the five estimates, rather than as evidence that the five cubes’ true effect sizes are identical – with only five studies, Q has limited power to detect real between-cube differences, and a non-significant result should not be over-interpreted as confirming homogeneity. Figure 1 shows all five estimates and the fixed-effect pooled result together.

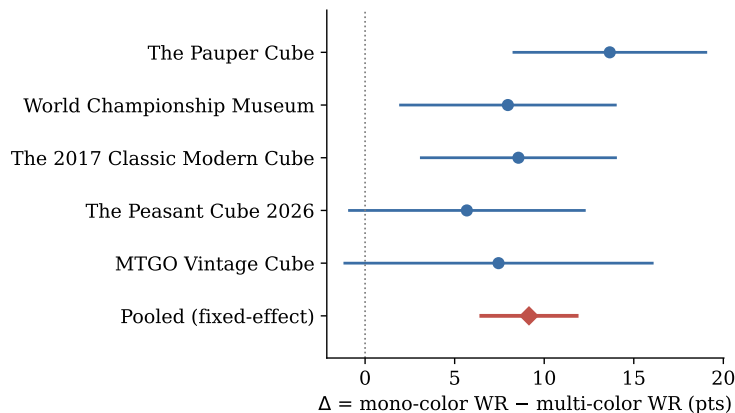


Figure 1: Draft-level Δ (mono-color WR – multi-color WR) per cube, 95% CI, and the fixed-effect pooled estimate across all five.

The overall picture we take from this section: a mono-color premium of roughly 6–14 percentage points at the individual-cube level, positive in direction in every cube tested, statistically robust to a conservative draft-level test in three of five individually, and – combined across all five via meta-analysis – clearly distinguishable from zero as a pooled effect even though two individual cubes did not reach significance alone. This is a more careful claim than “every cube shows a significant premium,” and we think it is the correct one to make; it is also, we think, a stronger one, since it does not depend on treating five moderate-sample estimates as if each had to clear a significance bar independently before the combined evidence could be taken seriously.

3.2 A complexity gradient: simple plans overperform, multi-step plans underperform

This is an exploratory finding, not a confirmatory one, and we present it as such. The “bottom 15” in each cube was identified purely by observed win rate; the characterization of what those cards have in common (cantrips, combo pieces, ramp) was produced afterward, by inspection, not

Table 4: Mean win rate of each cube’s 15 best- and 15 worst-performing reliable cards, and what mechanically characterizes the bottom 15 (author’s qualitative reading – see §2.3).

Cube	Top-15 avg	Bottom-15 avg	Spread	Bottom-15 characterized by
World Championship Museum	85.59%	23.15%	62.44pt	Cantrips & counterspells (Brainstorm, Opt, Serum Visions, Undermine, Mystical Tutor)
MTGO Vintage Cube	74.38%	17.87%	56.51pt	A near-complete storm-combo package (Dark Ritual, Doomsday, Tendrils of Agony, Yawgmoth’s Will, Underworld Breach)
The 2017 Classic Modern Cube	73.77%	24.60%	49.17pt	Artifact mana rocks & expensive colorless threats (Chromatic Sphere/Star, Hedron Archive, Blightsteel Colossus)
The Peasant Cube 2026	71.31%	27.18%	44.14pt	Graveyard reanimation & artifact ramp (Animate Dead, Exhume, Reanimate, Everflowing Chalice)
The Pauper Cube	69.79%	30.53%	39.26pt	Cantrips (Preordain, Behold the Multiverse, Growth Spiral, Capsize)

from a pre-defined or quantified complexity measure. Selecting extreme performers from a ranked list of hundreds of cards and then noticing a shared property is a standard and legitimate way to generate a hypothesis, but it is not yet a test of one – a reliability threshold of two distinct drafts (§2.3) is a meaningful floor against single-deck noise, but is weak enough that some cards in these tables may still reflect small-sample variance rather than the pattern described below.

With that caveat, the pattern is nonetheless consistent across all five cubes: the specific mechanic occupying the bottom of each cube’s card pool differs – unsurprisingly, since several of these mechanics (a full storm combo, artifact mana rocks) do not exist in every cube tested – but the *kind* of mechanic that ends up there does not. In every one of the five cubes, without exception, the worst-performing cluster is a strategy that only pays off after several correctly-sequenced, precisely-timed steps, while efficient, low-commitment creatures with an immediate board impact occupy the top of every cube’s list instead, regardless of that cube’s power level or design philosophy (Figure 2). The severity of the pattern also scales sensibly with how demanding each cube’s available multi-step strategies are: the MTGO Vintage Cube and World Championship Museum show the widest top-to-bottom spreads (56–62 points), while the Pauper Cube, which lacks true combo pieces, shows the narrowest (39 points) while still following the same direction. We treat this as a hypothesis worth a dedicated, quantified follow-up (§4.4) rather than as established here.

The same pipeline-stage confounding discussed for §3.1 in §4.1 applies here, and arguably more so: a bottom-15 cluster’s low win rate could reflect Forge’s match-play policy struggling to execute a multi-step plan, or it could equally reflect CubeCobra’s drafting/deck-construction bot failing to assemble a sufficiently coherent version of that plan in the first place (an incomplete storm package, a reanimation shell missing its enablers). Nothing in this section’s data distinguishes “the AI plays combo badly” from “the bot doesn’t draft combo well enough to play,” and we do not intend the

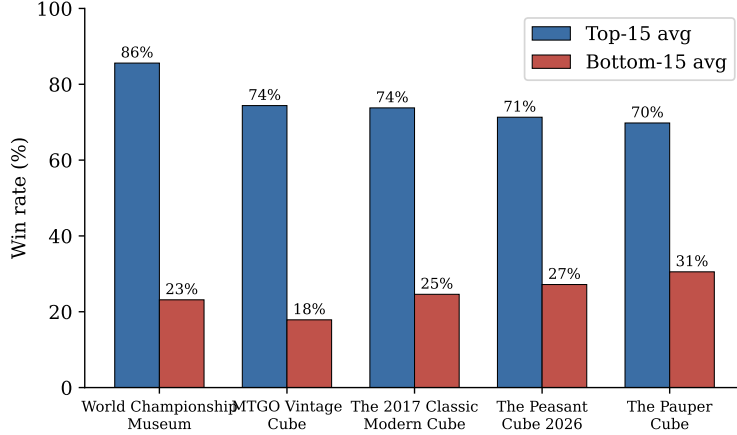


Figure 2: Top-15 vs. bottom-15 average reliable-card win rate, per cube.

phrase “complexity gradient” to imply we have.

3.3 Exploratory observation: the evasion keyword *Shadow* outperforms in two independently designed cubes

Two of the five cubes tested happen to include genuine creatures with the keyword *Shadow* (a static evasion ability under which the affected creature can be blocked only by other *Shadow* creatures) – the Pauper Cube and World Championship Museum. This is incidental to cube selection: the two cubes share essentially no other design overlap, sit at very different power levels, and were curated by different people.

Table 5: Win rate of every reliable *Shadow*-keyword card found across the five cubes tested.

Cube	Card	Win rate	Matches	Distinct drafts
The Pauper Cube	Dauthi Slayer	64.76%	105	15
The Pauper Cube	Soltari Trooper	62.34%	77	11
The Pauper Cube	Dauthi Horror	52.68%	112	16
World Championship Museum	Soltari Visionary	74.29%	35	5
World Championship Museum	Soltari Monk	71.43%	28	4
World Championship Museum	Soltari Priest	71.43%	42	6

Pooled across each cube’s three cards, *Shadow* averages 59.9% in the Pauper Cube (against that cube’s own 56.9% mono-color baseline) and 72.4% in World Championship Museum (against that cube’s own 53.9% mono-color baseline). We flag this explicitly as the most tentative finding in this paper: two cubes is a small basis for any general claim about a specific keyword, sample sizes per card are modest (4–16 distinct drafts), and we did not pre-specify *Shadow* as a mechanic of interest before observing these cubes’ card pools. We report it as an exploratory observation consistent with the complexity-gradient finding above – an always-on evasion ability requiring no activation or sequencing sits at the opposite end of the spectrum from §3.2’s worst performers – and as a candidate for confirmatory follow-up in a cube that includes it deliberately, not as a result this paper treats as established.

4 Discussion

4.1 What this study does and does not isolate

The central limitation of this study, stated plainly: **it cannot distinguish a bias originating in the drafting bot’s card evaluation, the same bot’s deck-construction logic, or Forge’s match-play policy.** All three are bundled into every result reported above, constant across all five cubes tested. It is entirely possible that CubeCobra’s bot simply drafts and builds better mono-color decks – tighter curves, more consistent mana, fewer speculative picks – and that Forge’s match-play engine is comparatively neutral; it is equally possible that deck construction is unbiased and Forge’s play policy favors simple game plans; either explanation, or some mixture of both, is compatible with the data reported here. This paper establishes that a large, consistent, direction-stable effect is observed under this shared pipeline across independently designed cubes; it does not establish that the pipeline causes that effect in any one of its specific stages, nor which stage would be responsible if it does.

Isolating the source would require a factorial design: holding the drafted card pool fixed while varying the deck-construction policy, and holding decks fixed while varying the match-play engine, then checking whether the effect tracks the drafting/deckbuilding stage, the match-play stage, or persists regardless. We did not attempt this here and present it as the natural next study.

4.2 A caution for cube designers using automated pipelines

A bot-draft-and-simulate pipeline is a genuinely useful tool for data-driven cube design, and nothing in this paper argues otherwise. But these results show that such a pipeline measures at least two things at once: the cube’s actual design, and systematic properties of the pipeline used to evaluate it. A single-cube study cannot cleanly separate the two without a point of comparison outside that one cube. A designer seeing their combo package or their card-selection spells underperform in simulation should weigh the possibility that the pipeline, not the card, is the limiting factor – particularly if the underperforming strategy requires several correctly-sequenced steps to pay off, which is precisely the pattern found here in five independently designed cubes.

4.3 Limitations

Pipeline-stage confounding. As discussed in §4.1, this study cannot determine whether observed effects originate in drafting, deck construction, or match simulation.

Non-independence of observations. Matches generated from the same draft share decks and are therefore not statistically independent; Table 3’s p -values should be read as suggestive rather than confirmatory. We addressed this directly with the draft-level paired test in Table 2, which is why that table, not Table 3, is this paper’s primary evidence for §3.1 – and why that more conservative test’s weaker result in two of the five cubes should be taken at face value rather than explained away.

Post-hoc characterization. The complexity-gradient finding (§3.2) was identified by inspecting results, not by testing a pre-defined, quantified hypothesis, and should be treated as exploratory.

Multiple comparisons. This paper reports several analyses over the same five-cube dataset without correcting for multiple comparisons; card-level findings in particular should not be read as confirmatory without independent replication.

Format. All matches are best-of-one: no card ever moves between a maindeck and a sideboard mid-match, so a card that only earns its keep in a specific matchup is played every game regardless

of opponent if maindecked. This could inflate the measured gap in §3.2 beyond what a format with real sidebar access would show; we do not have data here to estimate that effect’s size.

Scope. These results describe one specific software pipeline’s behavior on five card pools at one point in time; they are not a claim about human play, about any other simulation engine, or about how these cubes perform at an actual table.

4.4 Future work

Two follow-ups would meaningfully strengthen this line of inquiry – beyond the draft-level reanalysis already incorporated into §3.1 above. First, the factorial design outlined in §4.1 – varying the deck-construction and match-play stages independently – would localize the effects reported here to a specific pipeline stage rather than the pipeline as a whole. Second, a pre-registered, quantified complexity score (e.g. number of prerequisite cards, timing-sensitive interactions, or required resources for a strategy to function) tested directly against win rate via regression would convert §3.2 from a hypothesis-generating observation into a confirmatory test.

5 Conclusion

Running a bot-draft-and-simulate pipeline against five independently designed public cubes – spanning the power range from all-commons Pauper to Power-9 Vintage – reveals two directional patterns that recur across all five. A mono-color win-rate premium is positive in all five and, under a conservative draft-level paired test, statistically significant in three; the remaining two show the same direction (a majority of individual drafts favor mono-color) without reaching significance at this sample size, and we report that split honestly rather than rounding it up to “significant everywhere.” A much larger, more consistent gap between simple, low-commitment strategies and multi-step ones recurs without exception in all five, regardless of which specific mechanic occupies each end of that spectrum in a given cube. A more tentative, exploratory observation adds that where the evasion keyword Shadow happens to exist in two of the five, independently designed and otherwise unrelated, its cards outperform their own cube’s mono-color baseline in both. Taken together, this indicates that a meaningful share of the card-level performance signal produced by this pipeline is systematic across cubes, rather than attributable to any individual cube’s design alone. This study does not isolate which pipeline stage produces that property, and treats doing so as the natural next step – but the practical implication for cube designers holds regardless of that open question: a bot-draft-and-simulate pipeline should be treated as an instrument that may need calibration, not as a neutral measurement of a cube’s balance.

Data and Code Availability

All raw simulation logs, exported draft records (`cards.csv`, `SUMMARY.md`, `analysis.json`), and the pipeline scripts used to produce every number in this paper are available at *[GitHub repository URL to be added here]*. All statistics reported here were computed directly from those files by the scripts in that repository; none were transcribed or re-derived by hand.

References

1. CubeCobra. <https://cubecobra.com>. Bot-drafting simulator used to generate all draft data in this study.

2. Forge (Card Forge project). Open-source *Magic: The Gathering* rules engine, version 2.0.14. <https://github.com/Card-Forge/forge>.

Acknowledgments

All numerical results in this paper were independently recomputed from each cube’s raw `cards.csv` / `SUMMARY.md` export files. Cube selection was guided by CubeCobra’s own popularity and power-level categorization; card pools, archetype labels, and draft outcomes are each cube curator’s own and were not altered in any way for this study. Pipeline development and analysis were conducted through an iterative, multi-assistant process, including critical review of an earlier draft of this manuscript by multiple independent reviewers.